

Draft Guidance on Semaglutide

November 2023

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Active Ingredient: Semaglutide

Dosage Form: Solution

Route: Subcutaneous

Strengths: 2 mg/1.5 mL (1.34 mg/mL), 2 mg/3 mL (0.68 mg/mL), 4 mg/3 mL (1.34 mg/mL), 8 mg/3 mL (2.68 mg/mL)

Recommended Study: Request for waiver of in vivo bioequivalence study requirements

The recommendations and principles outlined in the most recent version of the FDA guidance for industry on *ANDAs for Certain Highly Purified Synthetic Peptide Drug Products That Refer to Listed Drugs of rDNA Origin*^a are applicable to this product.

Waiver request of in vivo testing:

To qualify for a waiver from submitting an in vivo bioequivalence study on the basis that bioequivalence is self-evident under 21 CFR 320.22(b), a generic semaglutide subcutaneous solution for injection product should be qualitatively (Q1)¹ and quantitatively (Q2)² the same as the reference listed drug (RLD).

¹ Q1 (Qualitative sameness) means that the test product uses the same inactive ingredient(s) as the reference product.

² Q2 (Quantitative sameness) means that concentrations of the inactive ingredient(s) used in the test product are within ±5% of those used in the reference product.

An applicant may seek approval of a drug product that differs from the RLD in preservative, buffer, or antioxidant if the applicant identifies and characterizes the differences and provides information demonstrating that the differences do not affect the safety or efficacy of the proposed drug product.³

Additional information:

Device:

The RLD is presented in a prefilled pen injector co-packaged with needles. The pen injector and needles are the device constituent parts.

FDA recommends that prospective applicants examine the size and shape, the external critical design attributes, and the external operating principles of the RLD device when designing the test device including:

- Multi-use disposable pen injector with fixed- or variable-dose format
- Dose selector and dose button
- Needle gauge and length

User interface assessment:

An abbreviated new drug application for this product should include complete comparative analyses so FDA can determine whether any differences in design for the user interface of the proposed generic product, as compared to the RLD, are acceptable and whether the product can be expected to have the same clinical effect and safety profile as the RLD when administered to patients under the conditions specified in the labeling. For additional information, refer to the most recent version of the FDA guidance for industry on *Comparative Analyses and Related Comparative Use Human Factors Studies for a Drug-Device Combination Product Submitted in an ANDA*.^a

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^a For the most recent version of a guidance, check the FDA guidance website at <https://www.fda.gov/regulatory-information/search-fda-guidance-documents>.

³ 21 CFR 314.94(a)(9)(iii).